

Impulsive millimetre flares from the planet-host M dwarf AU Microscopii at 106 GHz

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ABSTRACT

We present a twelve-execution ALMA Band 3 campaign on the planet-host M dwarf AU Microscopii. Two flares were caught: on 2025 January 19 a rise bracketed within 3.12 s across a gap in the recorded integrations (the intrinsic rise is unresolved) reached 29.4 ± 0.4 mJy with a 16-min e-fold and elevated emission for over 3 h (intermittently sampled); on 2025 May 26 a three-burst event reached 10.9 ± 0.5 mJy within 3 min (12-min e-fold). Band-3 radiated energies, $\geq 3.7 \times 10^{27}$ and $\geq 9.0 \times 10^{26}$ erg, are conservative lower limits of order 10^{27} erg (gaps not interpolated). Per-scan fluxes exceed the operational threshold for 12.7 per cent of the 6.18 h sampled on-source time (5.2 per cent of 501 scans) – a sampled-time activity fraction, not a duty cycle. The quiescent point source, 0.294 ± 0.017 mJy (campaign-average estimator; execution dispersion 0.032–0.035 mJy), agrees with the non-simultaneous 2018 12–25 GHz gyrosynchrotron extrapolation. Flux densities are parallel-hand $\text{Re}\langle XX \rangle$ (no full-Stokes calibration); Stokes- I conversions carry the model-dependent parallactic-angle bound (Section 2). A flat centimetre continuation is bounded at 0.34 mJy (95 per cent, parallel hand), with a model-dependent conservative Stokes- I bound of 0.54 mJy.

Key words: stars: flare — stars: individual: AU Microscopii — radio continuum: stars

1 INTRODUCTION

Magnetically active M dwarfs host quiescent gyrosynchrotron coronae punctuated by flares spanning many orders of magnitude in energy and duration (MacGregor et al. 2020; Tristan et al. 2026). AU Mic is a 22-Myr-old M1 dwarf at 9.725 pc hosting two confirmed transiting planets and a debated third candidate (Plavchan et al. 2020; Wittrock et al. 2022), an edge-on debris disk and frequent optical flaring (MacGregor et al. 2013; Gilbert et al. 2022). MacGregor et al. (2020) detected two ~ 35 -s strongly linearly polarised 220-GHz flares in archival 2015 ALMA data, interpreted as MeV-electron synchrotron. Tristan et al. (2026) decomposed the quiescent 12–25 GHz spectrum into a steep component ($\alpha = -0.88 \pm 0.10$, $S_\nu \propto \nu^\alpha$) and a flat component (0.64 ± 0.14 mJy) too bright for free-free emission, favouring optically thick gyroresonance.

No Band 3–6 millimetre monitoring existed between the 2015 archival events and this campaign. Twelve ALMA Band 3 executions (2025 January–June, program 2024.1.01095.S) anchor the quiescent spectrum between the centimetre and 220-GHz bands and catch two flares morphologically distinct from those 35-s events. We adopt $d = 9.725$ pc (Gaia), $P_{\text{rot}} = 4.8626 \pm 0.0013$ d (Gilbert et al. 2022), and the measured CHARA and SPIRou radii, $R_* = 0.75 \pm 0.03$ and $0.82 \pm 0.02 R_\odot$ (Plavchan et al. 2020; Donati

et al. 2023), quoted as a range wherever used (Supplementary Table S1.3).

2 OBSERVATIONS AND DATA REDUCTION

AU Mic was observed in twelve scheduling-block executions (EBs) from 2025 January 6 to June 3 in ALMA Band 3, 12-m array (three identical windows per execution, 5.4 GHz usable). The band spans 100.9–112.9 GHz, quoted as 106 GHz; the SED point of Section 3.1 uses the exact effective band centre, 105.6 GHz. Each execution has 4–62 target scans recorded as correlator dumps of 6.048 or 2.016 s (0.576-s integrations where retained); a flare *event* is defined in Section 3.4. In total, 6.18 h on source and 501 scans (execution log: Supplementary Table S1.2). We reduced the raw ASDMs in CASA 6.7.6 following the delivered QA2 recipes (detail in Supplementary S1), with phase-referenced per-scan photometry from the calibrated science-field visibilities, validated against TCLEAN and QA2 images ($\lesssim 10$ per cent).

All executions shared a J2000 phase centre (registration and proper-motion detail in Supplementary S1). The correlator recorded only the XX and YX products (the standard non-polarimetric continuum configuration), so linear polarisation cannot be measured; every flux density below is the parallel-hand quantity $\text{Re}\langle XX \rangle$; a dedicated full-polarisation campaign (per-cent-level $|Q/I|$ in one execution) is the decisive mechanism test (Section 4). The absolute flux scale is uncertain at 5 per cent, largely correlated within observing ses-

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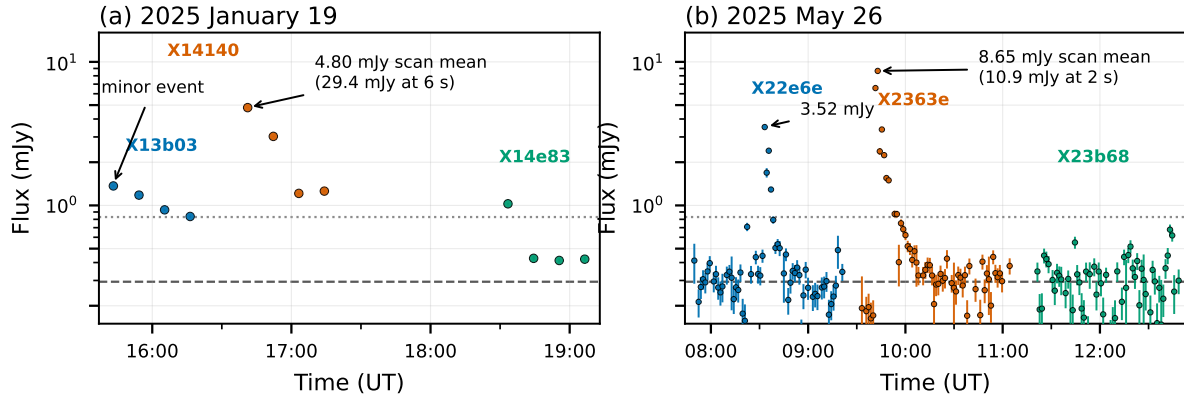


Figure 1. Band-3 per-scan light curve (1σ errors; floored at 0.15 mJy) for the only two epochs showing significant flaring activity – the remaining executions sit at quiescence (all 501 scans: Supplementary Table S1.1). The 11 scans (2.2 percent) consistent with zero or negative flux lie at or below the log-axis floor (signed values tabulated there). (a) 2025 January 19: three consecutive executions catch a minor excursion, the major flare (Fig. 3a), and the return to quiescence. (b) 2025 May 26: a four-scan excursion to 3.52 mJy (08:33 UT) and the impulsive flare (Fig. 3b). Dashed: quiescence, 0.294 mJy; dotted: the $5\sigma_{\text{MAD}}$ activity threshold, 0.83 mJy (Section 3.4).

sions and hence not folded into the morphology, onset, e-fold or rate uncertainties below; quoted uncertainties are statistical errors of the visibility mean, peaks carrying both explicitly. The scale was set per execution by the QA2 flux bootstrap against J1924–2914 (May–June; J2258–2758 in January, J1617–5848 for one January execution) with standard ALMA calibration-monitor models; cross-standard January–May comparisons conflate the two amplitude scales (Supplementary S1.6), execution-specific residuals enter the inter-execution scatter of Section 3.1, and the SPW-to-SPW gain is stable at $\simeq 2$ percent (0.25 systematic on in-band indices; Supplementary S1.6). Full correlator resolution was retained in the two flare executions (6.048- and 2.016-s dumps; 0.576 s in the January onset scan and burst 3): the pre-planned ASDM resolutions – ALMA standard observing has no real-time trigger – so the ≤ 3.12 -s onset brackets sample a fixed cadence, not a triggered response (Supplementary S1.2).

The parallel hand measures $I + Q'$ with $Q' = Q \cos 2\psi + U \sin 2\psi$ and ψ the parallactic angle; the unpolarised standards tie the campaign-mean scale to Stokes I , but an individual XX-derived flux can carry an unknown linear-polarisation contribution (cf. the strongly polarised 2015 events; MacGregor et al. 2020). Over the flare time-scales the effect is negligible – onset brackets, e-folds and in-band index ratios are polarisation-robust, and an L -level contribution to peaks and fluences is $\lesssim 1$ per cent. For the quiescent campaign average, a constant- I, Q, U model fit yields only an empirical bound from parallactic-angle modulation, $L < 0.19$ mJy (95 per cent) – a conservative bound under that stated model, not a calibrated polarisation measurement (geometry and the elevation degeneracy: Supplementary S1.6), propagated wherever a Stokes- I reading is required (Section 3.1). The cross-hand phase was not calibrated (no Stokes- V bound follows), and the YX series serves only as a null check against correlator artefacts (Supplementary S1.7).

3 RESULTS

3.1 Quiescent emission and the flat-component limit

We define flaring scans as per-scan means above an empirical $5\sigma_{\text{MAD}}$ activity threshold, median $+5\sigma_{\text{MAD}} = 0.83$ mJy ($\sigma_{\text{MAD}} = 1.4826 \times \text{MAD}$; dependences in Supplementary S1.1). The threshold is operational: defined after excluding the major-flare executions, it is not homogeneous across differing scan sensitivities (Supplementary S1.1), and the major flares exceed any plausible threshold by factors of ≥ 12 . The excluded sample removes the three January 19 executions (12 scans) and the May 26 execution from its onset onward (57 scans), leaving 432 scans; iterative 3σ clipping retains 404, whose median gives $S_q = 0.294$ mJy with a formal ± 0.017 mJy – the uncertainty of the campaign-average estimator (statistical, calibration and belt terms; execution-level 0.021–0.036 mJy, below; Supplementary S1.1). Inter-execution medians span 0.25–0.38 mJy ($\chi^2_\nu = 3.4$): a month-scale trend ($-11 \pm 4 \mu\text{Jy}$ per execution towards June, 2.9σ) plus $\simeq 32$ – $35 \mu\text{Jy}$ intrinsic execution-to-execution scatter, not a failure of the ± 0.017 measurement. The observations span ~ 30 stellar rotations, making a simple coherent rotational interpretation of the apparent monotonic trend unlikely, though the sparse sampling does not exclude rotationally modulated variability; nor is it a flux-scale drift: the QA2 bootstrapped transfer-calibrator fluxes bound any common drift over May–June at $\lesssim 1$ percent (Supplementary S1.6) – a possible long-term variation requiring monitoring. Carried into the campaign mean, these give the execution-level uncertainties quoted above; no conclusion depends materially on the choice (Supplementary S1.1). S_q is the campaign average of a slowly varying component (‘quiescent’ denotes this campaign-defined low state, not the absence of low-level activity); a 3700-K Rayleigh–Jeans (RJ) photosphere contributes only $16 \mu\text{Jy}$: the quiescent emission is dominated by non-photospheric stellar emission, $\sim 18\times$ brighter than the Rayleigh–Jeans photosphere.

The 9–40-au debris belt does not contaminate this photometry: a point-source-plus-annulus consistency check on the 48 retained quiescent scans of X2363e (window detail in Supple-

mentary S1.1) gives stellar flux 0.280 ± 0.020 mJy (annulus-fit belt 1.05 ± 0.34 mJy, $\chi^2_\nu = 1.1$; resolved-out ring-through -0.005 ± 0.007 mJy), while baselines beyond 150 k λ , which resolve the belt out, give 0.228 ± 0.025 mJy. The 52 ± 32 μ Jy (1.6σ) difference is not compelling, and the non-photospheric interpretation rests on the RJ comparison alone. The image-plane co-add, 0.317 mJy, reuses the same scans and calibration; the $\lesssim 0.09$ mJy spread across treatments tracks aperture and baseline choice (Supplementary S1.1).

The [Tristan et al. \(2026\)](#) steep component, propagated from its measured 2018 normalisation (0.56–2.36 mJy at 15 GHz, $\alpha = -0.88 \pm 0.10$) with no reference to our data, predicts 0.15–0.38 mJy (16–84 per cent) at 105.6 GHz, bracketing our measurement – consistency with the 2018 extrapolation, not evidence of persistence. At 222 GHz the same propagation (0.08–0.21 mJy) overlaps the B6 central fluxes ([Daley et al. 2019](#)) and the ARKS 245 ± 14 μ Jy central 1.35-mm flux ([Marino et al. 2026](#)), which resolves the belt, isolating the star (Fig. 2).

The flat component is bounded, as a model-dependent conservative Stokes- I upper bound, at ≤ 0.54 mJy (one-sided 95 per cent; 0.52–0.54 mJy across the error treatments above; a decomposition-independent envelope read as Stokes I with the model-dependent parallactic-angle polarisation bound $L < 0.19$ mJy of Section 2 admitted). The recorded parallel hand alone bounds it at 0.34 mJy (0.32–0.35 mJy), or 0.19–0.20 mJy (≤ 0.38 mJy in Stokes I) when the steep component is allowed its measured 2018 normalisation – a conditional, model-dependent constraint combining non-simultaneous measurements seven years apart (derivation and Monte Carlo draws in Supplementary S1.1). The unevolved 0.64 ± 0.14 mJy flat component of [Tristan et al. \(2026\)](#), if it remained spectrally flat to 106 GHz, exceeds the model-dependent conservative Stokes- I upper bound at only $\simeq 0.7\sigma$ – no significant inconsistency – while the parallel-hand bound gives 2.2σ at the execution-level $\sigma = 0.021$ mJy and 2.0σ at $\sigma = 0.036$ mJy (campaign and 2018 errors in quadrature); we treat the comparison as at most suggestive of inconsistency, admitting a 25–106-GHz turnover, a secular decline of $\gtrsim 40$ μ Jy yr $^{-1}$, or unresolved non-simultaneity over the ~ 7 -yr baseline. Optically thin free-free already required emission measures $\gtrsim 100\times$ wind models for the flat centimetre component ([Tristan et al. 2026](#)); our result is consistent with, but does not independently strengthen, that conclusion.

3.2 The 2025 January 19 flare

On 2025 January 19, three consecutive executions over 3.4 h caught a flare complex (Fig. 1a). X13b03 shows a minor excursion decaying from 1.37 to 0.84 mJy; X14140 (0.576-s dumps; Fig. 3a) resolves the onset: the source sits at quiescence (visibility-domain -0.8 ± 1.9 mJy) until 16:36:04 and the transition to the first flaring point, 21 ± 2 mJy, is bracketed within the following 3.12 s – a bracket across a $\simeq 2.5$ -s recording gap, not a measured rise time (Supplementary S1.2) – reaching the peak of 29.4 ± 0.4 mJy (statistical; ± 1.5 mJy calibration) at 16:36:35: $\sim 100\times$ the quiescent level. The peak is broadband (28.4–30.5 mJy), in-band $\alpha = -0.34 \pm 0.10$ (stat.) ± 0.25 (sys.) at ± 20 s, with no detected cross-hand signal ($\text{Im}\langle YX \rangle = -0.10 \pm 0.15$ mJy; Section 2). The decay has an e-fold $\tau_e = 16.1 \pm 1.1$ min (0–30 min post-peak), punctuated by 2–3.6-mJy excursions at

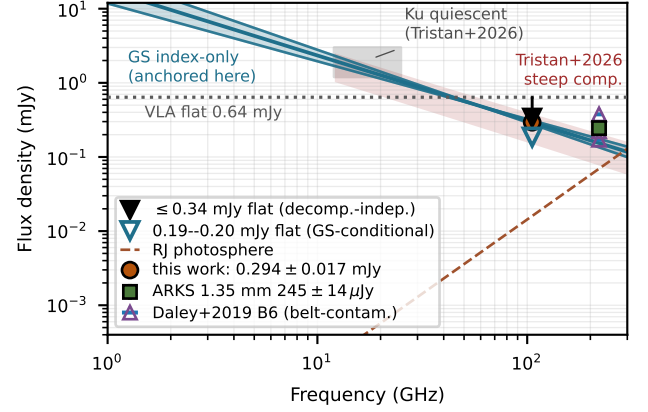


Figure 2. Quiescent centimetre-to-millimetre SED – *epochs non-simultaneous*, 2012–2025 (Section 3.1). Orange: this work, 105.6 GHz, 0.294 ± 0.017 mJy (formal uncertainty: statistical, calibration and belt terms; execution-level 0.021–0.036 mJy; parallel-hand-equivalent). Red band: the [Tristan et al. \(2026\)](#) steep component propagated from its measured 2018 normalisation (16–84 per cent range; Section 3.1). Blue band: index-only gyrosynchrotron extrapolations anchored to our measurement ($\alpha = -0.88 \pm 0.10$; [Tristan et al. 2026](#)); normalised to our point, they cannot independently test consistency. Green: ARKS 1.35-mm central flux ([Marino et al. 2026](#)); purple: B6 central fluxes ([Daley et al. 2019](#)) (belt-contaminated; B9 controls lie beyond the plotted range) – context only. Grey: 2018 Ku-band quiescent range ([Tristan et al. 2026](#)); dotted: their flat component. Triangles mark the flat-component bounds in the recorded parallel hand, distinguished by fill as well as colour: filled black, the decomposition-independent bound ≤ 0.34 mJy; open blue, the conditional constraint 0.19–0.20 mJy assuming the 2018 gyrosynchrotron normalisation (both 95 per cent; inflated by the model-dependent $L < 0.19$ mJy bound, these become ≤ 0.54 and ≤ 0.38 mJy as model-dependent Stokes- I bounds). Dashed: Rayleigh–Jeans photosphere.

22–34 min; X14e83 returns through 1.03 mJy to quiescence. The Band-3 radiated energy is

$$E_{\text{band}} = 4\pi d^2 \Delta\nu_{\text{eff}} \sum_i (S_i - S_q) \Delta t_i, \quad (1)$$

where the sum runs over retained on-source integrations at native cadence ($\Delta\nu_{\text{eff}} = 5.4$ GHz; S_i and S_q are parallel-hand flux densities, Section 2); recording gaps are not interpolated and negative excursions included – measured Band-3 lower limits, $\geq 3.7 \times 10^{27}$ erg over the January elevated state (≥ 204 min intermittent sampling), and $\geq 9.0 \times 10^{26}$ erg over the May 84 min (parameters: Supplementary Table S1.3).

Only ~ 4 per cent of the January energy is radiated in the onset bracket and initial peak – the decay carries 84 per cent, the tail 12 (Supplementary S1.2). The totals could be substantially larger for a synchrotron spectrum peaking outside the band or across the incompletely sampled January state. Relative to the quiescent band luminosity over the same recorded windows, the fluence ratios are $\simeq 1.7$ and 1.0.

3.3 The 2025 May 26 impulsive flare

Execution X2363e (09:33–11:04 UTC) is resolved by the 2.016-s dumps into *three* impulsive bursts within 3 min, followed by a smooth decay (Fig. 3b). The four-scan excursion

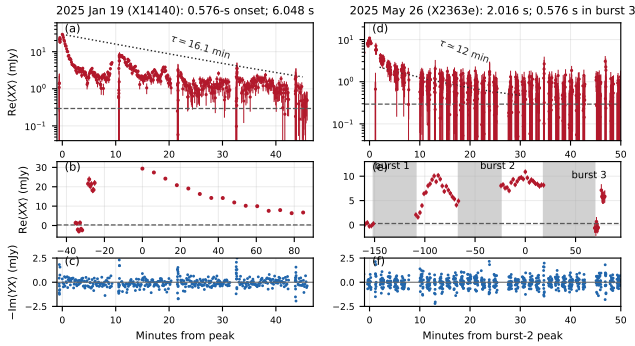


Figure 3. The two flares at full time resolution (dashed: quiescence). (a) 2025 January 19, onset scan (0.576-s dumps): quiescent through 16:36:04, then the transition to ~ 21 mJy bracketed within 3.12 s (across a recording gap) and peak 29.4 ± 0.4 mJy at +32 s. (b) 2025 May 26 (2.016-s dumps; 0.576 s in burst 3): the three impulsive bursts (shaded bands: scan gaps); burst 3 re-rises from quiescence to 7.1 mJy, bracketed within ≤ 3.12 s (same gap structure as panel a). Full executions and e-folds: Supplementary Table S1.3; cross-hand nulls: Section 2.

to 3.52 mJy 68 min before onset (Fig. 1b) is a separate minor event, exceeding the 15-min joining criterion (Section 3.4). The star was quiescent (0.25 ± 0.53 mJy) at 09:40:32.8; burst 1 is in progress by 09:41:15, so its onset is ≤ 42 s. Bursts peak at 09:41:37 (10.2 mJy), 09:43:04 (10.9 ± 0.5 mJy statistical, ± 0.5 mJy calibration, the event peak) and 09:44:20 (7.1 mJy), separated by 87 and 76 s, longer and brighter than the 2015 pair. The third burst, caught at 0.576-s cadence, re-rises from quiescence (-0.4 ± 0.6 mJy) to 7.1 mJy, the transition bracketed within ≤ 3.12 s – the same recording-gap bracket as January, not an independent coincidence (Supplementary S1.2).

The decay follows $\tau_e = 11.9 \pm 0.9$ min (2–20 min), reaching quiescence by ~ 21 min, with sporadic 1–3-mJy spikes to 82 min post-peak. It is bracketed by quiescent executions (12 min before onset; 1 h 35 min post-peak). The peak is $37 \times$ quiescence ($L_\nu = 1.2 \times 10^{15}$ erg s $^{-1}$ Hz $^{-1}$, $E_{\text{band}} = 9.0 \times 10^{26}$ erg; Eq. 1, same lower-limit caveats), broadband and mildly steep (in-band $\alpha = -0.71 \pm 0.52$ stat., ± 0.25 sys.), with no detected cross-hand signal ($\text{Im}\langle YX \rangle = -0.05 \pm 0.09$ mJy over ± 20 s). Neither epoch had contemporaneous optical/UV/X-ray coverage (TESS Sectors 1 and 27 cover 2018–2020; Sector 95 began seven weeks post-campaign).

The detection is stellar and survives every check: a point source at the stellar position in every baseline and window; onsets rising simultaneously in all windows and both antenna halves (calibration-independent baseline ratios; Supplementary Fig. S1.2); long baselines resolving out the belt; neighbouring executions at quiescence – location, not mechanism.

3.4 Flare occurrence and activity fraction

Over 6.18 h on source, per-scan fluxes exceed the operational 0.83-mJy threshold for 47.0 min – a sampled-time activity fraction of 12.7 per cent, not a duty cycle (threshold-robust; Supplementary S1.1). The scan fraction is smaller, 26 of 501 (5.2 per cent; Wilson 3.6–7.5 per cent), because elevated scans concentrate in the longer-scan flare executions; an *event* is consecutive above-threshold scans within 15 min

(the three May sub-bursts, 76–87 s apart, count as one). The two major flares give an observed rate of 0.32 h^{-1} above this threshold during the sampled ALMA integrations (Garwood Poisson 95 per cent $0.04\text{--}1.17 \text{ h}^{-1}$); the five minor excursions alone give an operational rate of 0.81 h^{-1} (95 per cent $0.26\text{--}1.89 \text{ h}^{-1}$; Supplementary Table S1.1), a distinct population not to be combined with the major-flare rate. The minor excursions are individually robust detections (each contains a scan $\geq 8\sigma$ above quiescence, systematic floor included) but, as unresolved one-to-four-scan excursions of 0.86–3.5 mJy, their astrophysical status is far weaker than the major flares’ (Supplementary S1.1). We therefore regard these five excursions as candidate astrophysical events rather than confirmed flares; the quoted rate is solely the occurrence rate of such above-threshold excursions under our operational definition. With $N = 2$ and 0.17 per cent of the 148-d span sampled, these are observed campaign rates, not the underlying rate: injection tests recover major-class events with unit probability, conditional on occurring during an execution, but only 4–25 per cent of 20-s threshold-level spikes (Supplementary S1.5). The MacGregor et al. (2020) 220-GHz rate (different band and cadence) is not directly comparable.

The archival controls corroborate the reduction (Section 2): constant Band-9 and Band-6 baselines with no flaring scan (January-class peaks extrapolate to 5.5–22 mJy, above every per-scan threshold; May-class to 2–8 mJy, marginal; Supplementary S2–S3); the same reduction run blind recovers the known 2015 June flare at 58σ , and comparison-star Band-7 fluxes are reproduced with no flare on 6-s to 40-min scales (Supplementary S4). These are empirical pipeline controls, not spectral constraints: a spectrum turning over between the bands would evade such non-detections.

4 DISCUSSION AND CONCLUSIONS

If the emitting plasma covers a stellar disc, the peaks reach brightness-temperature lower limits of $\geq (7.4, 2.8) \times 10^6$ K (January, May) at the SPIRou radius, 19 per cent higher at the CHARA radius (Supplementary Table S1.3) – not diagnostic: $T_b \propto (R_*/r)^2$ rises steeply for a smaller emitting radius (illustratively $\sim 7 \times 10^8$ K at $r \sim 0.1 R_*$). The ≤ 3.12 -s onset brackets, if they trace light travel, confine the emitting region to $\lesssim ct \approx 9.4 \times 10^{10}$ cm, $1.64\text{--}1.79 R_*$ for the measured radii – an illustrative causality scale, conditional on light-crossing governing the transition, not a measured or observational size limit; beaming or gradual acceleration would relax it. Neither transient is associated with the extended debris belt: both flares are unresolved in the *uv* plane, the amplitude dropping only ~ 20 and ~ 15 per cent to the longest baselines (Supplementary S1.3): Gaussian-equivalent scales $\theta \simeq 0.24$ and $\simeq 0.10$ arcsec ($\simeq 2.3$ and $\simeq 1.0$ au at 9.725 pc), the circular-Gaussian FWHM matching the observed amplitude ratios, not confidence limits. They establish a compact transient at the stellar position, inconsistent with emission distributed over the 9–40-au belt (stellar angular radius ≈ 0.4 mas); they do not resolve a stellar magnetosphere – location and rapid variability favour a stellar origin.

The observed broadband emission is compatible with incoherent gyrosynchrotron emission – plausible, but not uniquely established by these data: neither full-Stokes polarisation nor contemporaneous optical/UV/X-ray coverage exists for ei-

ther event. The $|Q/I| > 0.12$ signature by which MacGregor et al. (2020) interpreted their 2015 spikes as linearly polarised MeV-electron synchrotron is undetectable here (Section 2; Supplementary S1.6). Brightening at 106 GHz requires either gyrofrequency harmonics $s \sim 10\text{--}20$ in kG-level fields – consistent with AU Mic’s measured 2.1–2.3-kG mean field (Kochukhov & Reiners 2020) and ~ 4 -kG small-scale fields (Donati et al. 2025) – or multi-MeV electrons in a few-hundred-gauss corona (MacGregor et al. 2020); index-based inferences, including the ultrarelativistic $p = 1 - 2\alpha$ relation, are illustrative only (Supplementary S1.4).

The in-band index evolves through the January decay, from $\alpha = -0.86 \pm 0.37$ at 1–5 min post-peak to $+0.96 \pm 0.54$ at 20–45 min; the May event drifts similarly, from -0.71 ± 0.52 at peak to $+1.35 \pm 1.04$ at 10–25 min. The nominal January trend (3.0σ under statistical errors) is estimator-dependent and does not survive the strictest tests: it falls below 2σ omitting the most influential window, and a window-mean SPW estimator gives no trend (permutation floor $p = 1/64$; systematics propagation in Supplementary S1.4). We present the apparent thin-to-thick progression as an indication requiring confirmation only – it is not used as evidence for the emission mechanism or against homogeneous expansion.

The May burst train shows richer structure: three bursts separated by 87 and 76 s could arise from successive magnetic energy-release and particle-injection episodes superposed on a decaying trapped-electron reservoir – a hypothesis, not a demonstration, and distinct from the ~ 35 -s spike pairs of MacGregor et al. (2020); the January excursions need not share the process. ~ 96 per cent of the January fluence is radiated after the initial onset bracket and peak, in the extended decay and elevated-state tail – with two events, these associations describe these events, not the population.

Within the limited published AU Mic and Proxima Centauri ALMA compilation (Supplementary Table S1.5), the January event has the highest reported peak spectral luminosity, $3.3 \times 10^{15} \text{ erg s}^{-1} \text{ Hz}^{-1}$ – $\sim 1.7 \times$ AU Mic’s own brightest 2015 event (MacGregor et al. 2020) and $\sim 14 \times$ the brightest of the 50-h Proxima Centauri campaign (Burton et al. 2025). This is a ranking of non-simultaneous events at unmatched cadences and frequencies – observational context, not a claim of energetic superiority (L_ν at 106, 220 and 233 GHz is not automatically comparable for a self-absorbed spectrum with frequency-dependent turnover). Binned to the 2015 35-s cadence the January peak averages 27.0 ± 0.2 mJy, still $1.6 \times$ that event – not a finer-sampling artefact.

We conclude in three tiers of security. *Secure*: two Band-3 flares – compact transients at the stellar position – with peak fluxes 29.4 ± 0.4 and 10.9 ± 0.5 mJy, transitions from the last recorded quiescent point to the first flaring point bracketed within 3.12 s across gaps in the recorded integrations (the intrinsic rise times are unresolved), 12–16-min e-folds, elevated emission across 1–3-h elapsed spans (intermittently sampled), measured Band-3 radiated energies over the recorded intervals $\geq 3.7 \times 10^{27}$ and $\geq 9.0 \times 10^{26}$ erg (lower limits), and a quiescent 106-GHz point source at 0.294 ± 0.017 mJy (formal campaign-average uncertainty; execution-level variability and systematics imply $0.021\text{--}0.036$ mJy for between-epoch comparisons). *Moderately constrained*: a compact origin at the stellar position inconsistent with the 9–40-au belt; a model-dependent conservative Stokes- I flat-component bound ≤ 0.54 mJy (95 per cent, via the parallactic-

angle polarisation bound of Section 2; the recorded parallel hand bounds it at 0.34 mJy, or 0.19–0.20 mJy conditional on the steep component retaining its 2018 normalisation, a 25–106-GHz turnover or a decline since 2018 remaining open; Section 3.1); and the observed campaign rates during the sampled integrations, 0.32 h^{-1} (two major events) and 0.81 h^{-1} (five minor excursions; an operational rate, not to be combined with the major-flare rate), with a sampled-time activity fraction of 12.7 per cent (per-scan fluxes above the operational threshold over the 6.18 h on source). *Tentative*: the gyrosynchrotron interpretation and the spectral-index evolution, which does not survive the strictest estimator test and is not used here as evidence against homogeneous expansion. Such variability also contaminates single-epoch millimetre photometry: the central flux can be elevated by factors of tens during flares, with lower-level enhanced emission persisting for hours; fast-spike flagging (cf. Marino et al. 2026) would not remove a January-type state, and residual central emission in deep maps requires modelling the unresolved stellar component. Simultaneous ALMA, optical/UV and X-ray monitoring is required to connect the millimetre flare emission to the thermal flare energy budget.

ACKNOWLEDGEMENTS

The archival projects used are listed in Supplementary Sections S2–S3; this paper also uses ALMA data ADS/JAO.ALMA/#2024.1.01095.S.

AUTHOR CONTRIBUTIONS

The data-reduction/analysis pipeline used the Python-based framework described by White (2026, MNRAS RASTI-26-047, submitted), which incorporates AI-assisted software-development and validation tools. The scientific content, analysis, and conclusions were designed, executed, and verified by the author. All errors and omissions are the responsibility of the author, who declares no conflict of interest.

DATA AVAILABILITY

The ALMA data are public in the ALMA Archive; per-scan photometry, light curves, spectral indices, control analyses and regenerating scripts are in Supplementary Material (Zenodo DOI pending).

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